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Experimental and theoretical studies on decomposition of pyrrolidine

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Abstract

A laminar premixed pyrrolidine/oxygen/argon flame with volume ratio C/O/N of 1.0/2.4/0.25 at 40 mbar was investigated using molecular-beam mass spectrometry (MBMS) with tunable synchrotron vacuum ultraviolet (VUV) photoionization. About 50 species were identified from photoionization efficiency (PIE) spectra, including various air pollutants such as nitric oxide, ammonia, hydrogen cyanide, acetonitrile, formaldehyde, acetaldehyde, and propanal. Some radicals were detected including C_4H_8N, C_4H_7, C_3H_5, C_3H_3, C_2H_5, CHO, and CH_3. Detected molecular intermediates include the isomers of C_2H_4O (ethenol and acetaldehyde), C_3H_4 (propyne and allene), C_4H_6 (1,3-butadiene and 1-butyne) and C_4H_8 (1-butene and 2-butene). Based on the experimental observations, the decomposition and ring-opening pathways of pyrrolidine were calculated at G3B3 level. Also, likely reaction channels from the products of H- and H_2-elimination reactions were proposed. This combined experimental and theoretical investigation intends to shed light on the decomposition channels of pyrrolidine which may assist the development of a kinetic model for pyrrolidine pyrolysis and combustion.
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Keywords: Pyrrolidine; Low pressure premixed flame; Decomposition pathway; G3B3

1. Introduction

Combustion of nitrogenous fuels may contribute to the emission of NO_x [1], which can cause the formation of acid rain, photochemical smog and ground-level ozone. To assist elucidating the formation of NO_x from pyrolysis and combustion of complex fuels such as coal or biomass, one strategy is to use model compounds with typical chemical features such as heterocyclic rings. N-containing monocyclic model compounds investigated to date include pyrrole, pyridine, 2-picoline and pyrrolidine [2]. Extensive studies of the pyrolysis, oxidation and combustion of pyrrole [3–10] and pyridine [11–18] have been performed; both are aromatic compounds with unsaturated bonds. In contrast, information on the pyrolysis and

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combustion of pyrrolidine, a cyclic amine featuring a five-membered saturated ring, is limited. Pyrrolidine is found in leaves of several plants and it represents a structural element of the amino acids proline and hydroxyproline. Structural motifs of nitrogenated fuels may influence the NO_x formation mechanism and the distribution of major and minor species, making pyrrolidine an interesting model compound for further study.

In 1987, Lifshitz et al. investigated the pyrolysis of pyrrolidine in single-pulse shock tube experiments in the temperature range of 900–1400 K and at densities of $\sim 3 \times 10^{-5}$ mol/cm^3 [19]. About 15 stable species were observed with gas chromatography (GC) and GC/MS methods. They proposed $CH_2CH_2 + (CH_2)_2NH$ and $CH_3CHCH_2 + CH_2NH$ as initial reactions of pyrrolidine pyrolysis via direct ring cleavage. Recently, the pyrolysis and combustion of pyrrole and pyridine was investigated in detail [9,10,17,18] with synchrotron-based photoionization molecular-beam mass spectrometry (MBMS).

In continuation of this work, a slightly fuel-rich premixed pyrrolidine flame at 40 mbar is investigated here to identify intermediates formed in the flame. Furthermore, the decomposition and ring-opening pathways of pyrrolidine are calculated at the G3B3 level, taking the experimental observations into account. The calculations include further reaction channels from the products of the major H- and H_2-elimination reactions.

2. Experimental

The experimental work was performed at National Synchrotron Radiation Laboratory (NSRL), Hefei, China. The instrument was described in detail elsewhere [20]. It consists of a flame chamber containing a moveable 6 cm diameter McKenna burner, a differentially-pumped chamber with a molecular beam sampling system, and a photoionization chamber combined with a reflectron time-of-flight mass spectrometer (RTOF-MS). The combustion species were sampled by a quartz cone with a 40° included angle and a ~500 μm orifice at the tip. The formed molecular beam passed through a nickel skimmer into the photoionization chamber, where it was crossed by the synchrotron light, and then the photoions were collected and analyzed by RTOF-MS.

The pyrrolidine flame at 40 mbar was slightly fuel-rich (C/O/N = 1.0/2.4/0.25). The flow rates of O_2 and argon were 1.242 and 0.500 standard liter per minute (SLM), respectively. The flow rate of liquid pyrrolidine was 0.955 ml/min (equal to 0.258 SLM in the gas phase) and was controlled by a syringe pump. The inlet mass flow rate was 2.055×10^{-3} g s^{-1} cm^{-2}, and the cold-flow (273 K) velocity was 29.87 cm s^{-1}. A series of mass spectra was taken in the center of the luminous region, varying the photon energy. The integrated ion signals for a specific mass were normalized by the photon flux and plotted as a function of photon energy to produce photoionization efficiency (PIE) spectra, which contain precise information on the ionization energies (IEs) of the corresponding species. Considering the limited cooling effect of the molecular beam [21], the experimental error for these determined IEs is within 0.05 eV for species with strong signal-to-noise ratios (SN) and may be slightly higher for some weak species.

3. Computational method

The highly precise G3B3 calculation method was employed to study the decomposition pathways of pyrrolidine. The G3B3 method is a variation of Gaussian-3 theory in which the geometries and frequencies were calculated at the B3LYP/6-31G(d) level. It includes a series of single-point energy calculations with the QCISD(T,E4T)/6-31G(d), MP4/6-31+G(d), MP4/6-31G(2df,p), and MP2(Full)/G3large methods based on the optimized geometries at the B3LYP/6-31G(d) level. Zero-point energies (ZPE) were obtained from the frequency calculations and scaled by a factor of 0.96. Higher-level corrections (HLC) and spin–orbit corrections (ΔE(SO), only for atomic species) were also included. Then the G3B3 energies were obtained. IRC analysis was carried out to confirm the transition states connecting both the reactants and products. For reactions including biradicals, we calculated the energies at CASPT2/6-31G(d)//CAS(6,6)/6-31G(d) level [22,23], which are provided in the Supplementary data as Tables S1 and S2. The details of the G3B3 theory have been described in the literature [24,25], and all calculations were performed with the Gaussian 03 program [26].

4. Results and discussion

4.1. Photoionization mass spectra

Fig. 1 shows four photoionization mass spectra taken at a sampling position of 2.5 mm from the burner, in the center of the luminous region, with photon energies of 9.0, 10.5, 11.7, and 14.2 eV, respectively. Most species have high signals at this position. The number of peaks changes with photon energy. At 9.0 eV, only few signals are observed at a mass m to charge z ratio m/z of 41, 43, 57, and 67–71 together with some species heavier than mass 71. At 10.5 eV, a number of combustion intermediates are detected with peaks at m/z 15, 17, 28–30, 40–44, 50, 52, 54–58, 78–85, and 92. A noteworthy feature in the mass spectrum

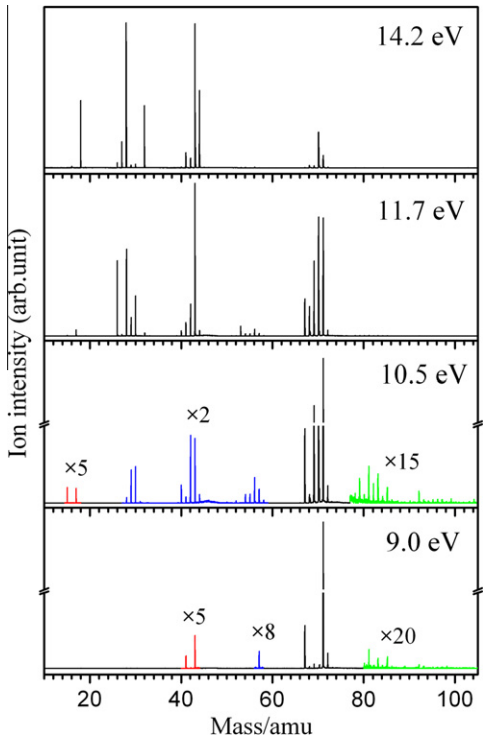

Fig. 1. Photoionization mass spectra of the pyrrolidine flame taken at 2.5 mm from the burner with four different photon energies labeled in the figure.

at 11.7 eV is the high signal at m/z 43 ($C_2H_5N^+$), caused by photodissociation of pyrrolidine. In the mass spectrum at 14.2 eV, the signatures of major species are detected, including H_2O, CO, O_2, and CO_2. The observed signals in the range of m/z 2 to 99 correspond to hydrocarbon, oxygenated, and nitrogenous species. They can be combustion intermediates or fragment ions from photoionization. Assignment of these species, which can be distinguished by different IEs, is performed with the aid of PIE spectra.

4.2. Identification of combustion intermediates

Table 1 lists the species identified in this work, comparing the measured IEs with data from literature [27]. For species with previously unreported values, IEs were calculated at the G3B3 level. Fig. 2 illustrates PIE spectra of four nitrogenous species at m/z 27, 29, 30, and 53 observed in the flame. For m/z 27 (Fig. 2a), the onset is 13.62 eV, which corresponds to the IE of hydrogen cyanide (IE = 13.60 eV [27]). Fig. 2b shows PIE of m/z 29 with the onset located at 9.96 eV, which implies methanimine formed in the flame (IE = 9.97 eV [27]). Two onsets are observed at 9.25 and 10.87 eV for m/z 30 (Fig. 2c) which correspond to ionization of nitric oxide (IE = 9.26 eV [27]) and formaldehyde (IE = 10.88 eV [27]), respectively. For m/z 53, the onset is 10.92 eV (Fig. 2d), corresponding to 2-propenenitrile (IE = 10.91 eV [27]).

Other N-containing species were detected and tentatively assigned by their m/z ratios and IEs (see Table 1), including ammonia, nitrogen, methylamine, acetonitrile, ethenamine, CHNO, propiolonitrile, 2-propen-1-imine, N-ethylidenemethylamine, pyrrole, C_4H_7N and C_4H_8N with masses below 71, and larger species such as pyridine, 2-methylpyrrole, and piperidine. For acetonitrile and CHNO, even though IEs are not indicated because the onsets are obscured by the pyrrolidine fragment at m/z 41 and 43, we have identified them by high-resolution mass spectrometry with electron ionization in Bielefeld. Also, CH_3CN was previously observed in the pyrolysis of pyrrolidine [19]. For CHNO, isocyanic acid (HNCO) is the more favorable isomer in the flame, owing to its stability (according to Gaussian-2 quantum chemical calculations [28]). It had also been identified in flames of other N-containing fuels [9,17,29]; fulminic acid (HCNO) cannot be excluded, however, according to previous reports [17,30]. Furthermore, C_4H_7N may contain both isomers, 3,4-dihydro-pyrrole and 2,3-dihydro-pyrrole.

Several oxygenated species were also identified including formaldehyde, methanol, ketene, acetaldehyde, and propanal. Ketene is a common intermediate in hydrocarbon flames which has also been observed in flames of other nitrogenous fuels [9,17,29]. Formaldehyde, acetaldehyde, and propanal as harmful air pollutants deserve special attention [31]. We also detected ethenol, which has been identified as an intermediate in flames of various hydrocarbons [32,33], oxygenated [34], and N-containing fuels [9,17,29,30].

Besides nitrogenated and oxygenated species, hydrocarbon intermediates are involved in the reaction mechanism of N-containing fuels. In the pyrrolidine flame, C1–C7 hydrocarbons were identified. C_3H_4 is composed of both allene and propyne in accord with observations from other N-containing flames [9,17]. C_4H_2 and C_4H_4 are identified as 1,3-butadiyne and vinylacetylene, respectively. C_4H_6 contains both 1,3-butadiene and 1-butyne. Also, C_4H_8 represents both 1-butene and 2-butene. Further species at m/z 68, 82, 83, 84 and heavier than 92 were observed but did not present clear onsets in the PIE spectra, precluding unambiguous assignment.

4.3. Decomposition pathways of pyrrolidine

Fuel decomposition reactions in flames include both pyrolysis and H-abstraction upon radical attack (R = H, O, and OH). The latter type is a most probable starting reaction in flames of saturated hydrocarbons and their derivatives, includ-

Table 1
Combustion intermediates measured in the premixed pyrrolidine/oxygen flame.

Mass	Formula	Name	IE (eV)		Mass	Formula	Name	IE (eV)	
			This work	Literature[a]				This work	Literature[a]
15	CH_3	Methyl radical	9.83	9.84	50	C_4H_2	1,3-Butadiyne	10.16	10.17
16	CH_4	Methane	12.60	12.61	51	C_3HN	Propiolonitrile	11.59	11.62
17	NH_3	Ammonia	10.08	10.07	52	C_4H_4	Vinylacetylene	9.57	9.58
26	C_2H_2	Acetylene	11.37	11.40	53	C_3H_3N	2-Propenenitrile	10.92	10.91
27	HCN	Hydrogen cyanide	13.62	13.60	54	C_4H_6	1,3-Butadiene	9.08	9.07
28	C_2H_4	Ethene	10.50	10.51		C_4H_6	1-Butyne	10.20	10.18
29	CH_2NH	Methanimine	9.96	9.97	55	C_3H_5N	2-Propen-1-imine	9.58	9.65
	C_2H_5	Ethyl radical	<8.4	8.12		C_4H_7	1-Buten-3-yl radical	7.50	7.49
	CHO	Formyl radical	<8.4	8.12	56	C_4H_8	1-Butene	9.60	9.55
30	NO	Nitric oxide	9.25	9.26		C_4H_8	2-Butene	9.12	9.11
	CH_2O	Formaldehyde	10.87	10.88	57	C_3H_7N	N-Ethylidenemethylamine	7.75	7.70[b]
31	CH_3NH_2	Methylamine	8.98	8.90	58	C_3H_6O	Propanal	9.95	9.96
32	CH_3OH	Methanol	10.83	10.84	66	C_5H_6	1,3-Cyclopentadiene	8.52	8.57
39	C_3H_3	Propargyl radical	8.70	8.67	67	C_4H_5N	Pyrrole	8.20	8.21
40	C_3H_4	Allene	9.75	9.69	69	C_4H_7N	3,4-Dihydro-pyrrole	9.32	9.26[b]
	C_3H_4	Propyne	10.37	10.36		C_4H_7N	2,3-Dihydro-pyrrole	7.42	7.26[b]
41	CH_3CN	Acetonitrile[c]	–	12.20	70	C_4H_8N	Pyrrolidine radical[c]	–	–
	C_3H_5	Allyl radical	8.10	8.10	71	C_4H_9N	Pyrrolidine	8.40	8.41
42	C_2H_2O	Ketene	9.61	9.62	78	C_6H_6	Benzene	9.25	9.24
	C_3H_6	Propene	9.79	9.73	79	C_5H_5N	Pyridine	9.32	9.26
43	C_2H_5N	Ethenamine	8.18	8.20	81	C_5H_7N	2-Methylpyrrole	7.77	7.78
44	C_2H_4O	Acetaldehyde	10.24	10.23	85	$C_5H_{11}N$	Piperidine	8.05	8.03
	C_2H_4O	Ethenol	9.32	9.33	92	C_7H_8	Toluene	8.82	8.83

Note: IE error for high SN is ±0.05 eV, for weak SN is ±0.1 eV in this work.
[a] Refer to [25] except for specific description.
[b] Refer to calculation using the G3B3 method.
[c] Detected by high-resolution mass spectrometer.

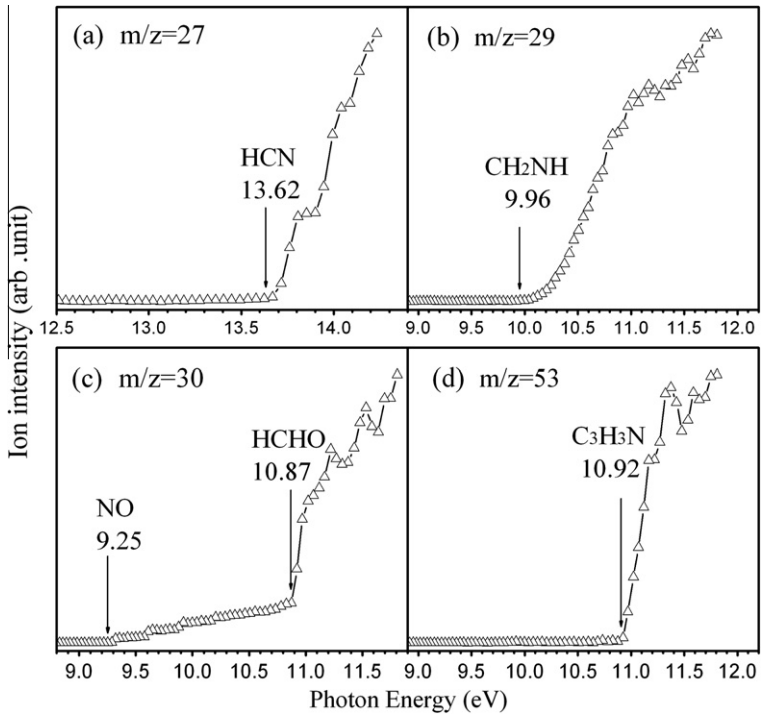

Fig. 2. PIE spectra of (a) m/z 27 (HCN), (b) m/z 29 (CH_2NH), (c) m/z 30 (NO/CH_2O), and m/z 53 (C_3H_3N) measured in the flame at 2.5 mm from the burner.

ing morpholine [29] and cyclohexane flames [35]. The main reactions in the pyrrolidine flame include decomposition via radical attack and the subsequent β-scissions pathways. Bond dissociation energies (BDEs) for the α-C–H, β-C–H and N–H bonds can be evaluated from the calculations, as shown in Fig. 3. Cleavage of the α-C–H and N–H bonds is easier than that of the β-C–H and accordingly, the barriers of the H-abstraction reactions of α-C–H and N–H are lower. The pyrrolidine radical has been detected and would likely be **PC1** or **PC2** (Fig. 3). Ring-opening from pyrrolidine radical can occur via β-scission, and subsequent decomposition products include CH_2CH_2, CH_2NH, CH_2CHCH_2, CH_2NCH_2, and CH_2CHNH. Some of these species were detected in the experiment. Due to the length limitations of the paper, we do not discuss the radical attack reactions in detail.

The main direct ring-opening and decomposition pathways of pyrrolidine were calculated at the G3B3 level. Also, the reaction channels starting from the H-elimination products were computed. The ring-opening products can easily decompose to small molecular products, and their subsequent reaction pathways were not included in the calculations. Calculation results are shown in Figs. 3–5 and in the Supplementary data.

4.3.1. Decomposition and isomerization pathways of pyrrolidine

The major ring-opening and H-elimination pathways of pyrrolidine are shown in Fig. 3. Ring opening can occur via intramolecular hydrogen

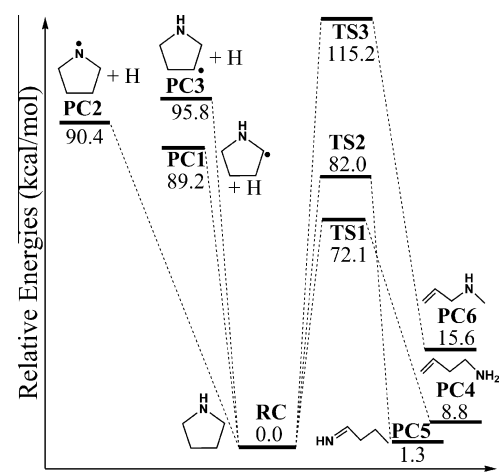

Fig. 3. H-elimination and isomerization reactions of pyrrolidine.

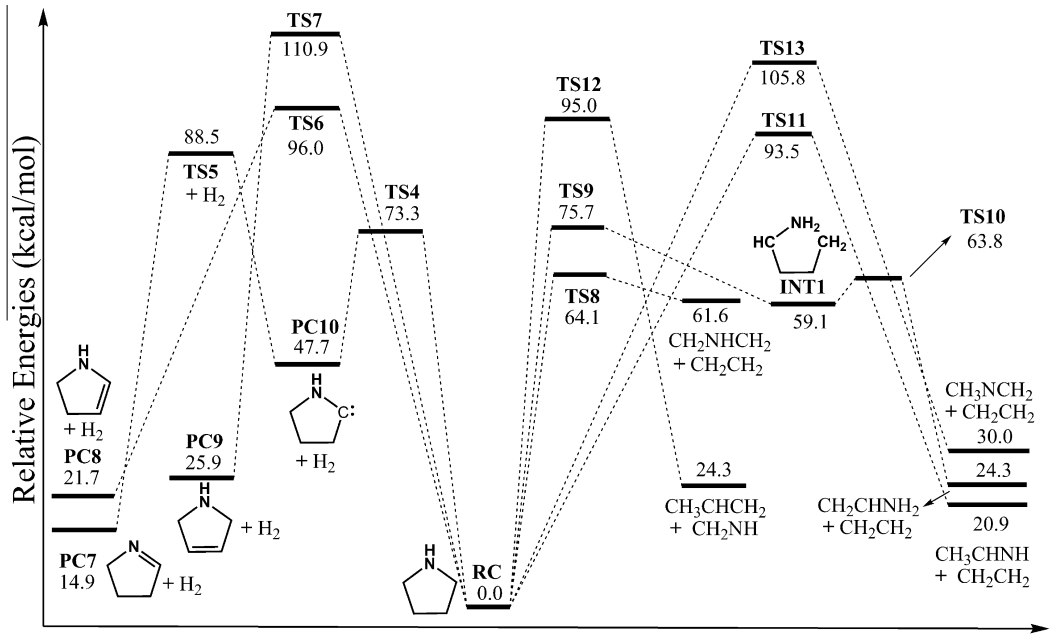

Fig. 4. H_2-elimination and decomposition reactions of pyrrolidine.

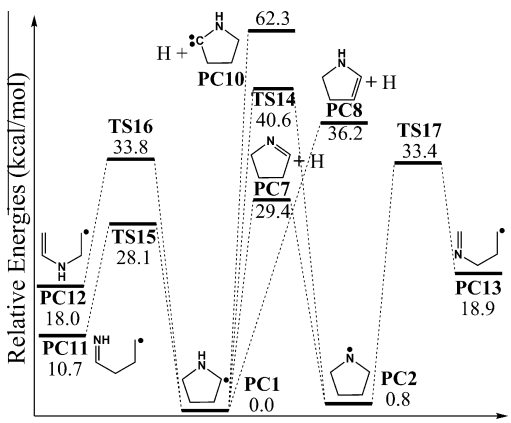

Fig. 5. Isomerization and decomposition reactions of **PC1** and **PC2** (energies are relative to **PC1**).

transfer and bond cleavage. The barrier of α-C–N bond cleavage via hydrogen transfer from β-C to N is lowest, with 72.1 kcal/mol (**TS1**). The barrier of α-C–N bond cleavage via hydrogen transfer between two α-C's is 82.0 kcal/mol, while the barrier is relatively lower (71.5 kcal/mol) at CASPT2 level (see Table S1). The calculation results have some difference between the multireference and G3B3 methods for the biradicals. However, the product **PC5** is more stable than **PC4**. Obviously, the barrier of α-C–β-C bond cleavage is higher (115.2 kcal/mol). The ring-opening products can further decompose to smaller molecular (or radical) products.

The direct cleavage pathways of pyrrolidine include the C–H and N–H bond dissociation reactions to form pyrrolidine radicals + H, similar to the H-abstraction pathways. The cleavage of the α-C–H and N–H bonds is easier and the dissociation energies to form **PC1**, **PC2**, and **PC3** are 89.2, 90.4, and 95.8 kcal/mol, respectively. Subsequent reaction pathways of **PC1** and **PC2** are discussed in Section 4.3.2.

In addition to H-elimination and ring-opening reactions, pyrrolidine can decompose to some small molecular products such as H_2 and ethene (CH_2CH_2). Dihydro-pyrrole can be formed via H_2-elimination reactions. Fig. 4 shows that the formation of 3,4-dihydro-pyrrole (**PC7**) is initiated by H_2-elimination from the α-C atom to a carbene product **PC10**, which subsequently transfers hydrogen to **PC7**. The second step requires a greater overall barrier, with a transition state (**TS5**) lying 88.5 kcal/mol above pyrrolidine. For **PC10**, it was optimized as a singlet. The 2,3- and 3,4-hydrogen elimination can form 2,3-dihydro-pyrrole (**PC8**) + H_2 and 2,5-dihydro-pyrrole (**PC9**) + H_2, with barriers of 96.0 and 110.9 kcal/mol, respectively. Thus the **PC7** + H_2 formation pathway is the lowest-energy H_2-elimination channel. The stable dihydro-pyrrole product (**PC7**) has been detected in the experiment.

Five possible decomposition pathways of pyrrolidine to smaller molecular products via ring-opening reactions are shown in Fig. 4. (**a**) Decomposition to $CH_2CH_2 + CH_2NHCH_2$ can occur via two α-C–β-C bond cleavages with an

energy barrier of 64.1 kcal/mol (**TS8**), which is consistent with the calculation result by Ess and Houk using the CBS-QB3 method [36]. However, CH_2NHCH_2 was not detected here. The reason may be the high formation enthalpy of CH_2NHCH_2, which can readily isomerize. (**b**) The products $CH_2CH_2 + CH_2CHNH_2$ are formed via two transition states and one well. The ring-opening intermediate (**INT1**) is first formed via the hydrogen transfer reaction as the rate-determining step, with a barrier of 75.7 kcal/mol. The barrier of the subsequent dissociation reaction is only 4.7 kcal/mol above **INT1**. For pathways (**c**) $CH_2CH_2 + CH_3CHNH$ and (**d**) $CH_2CH_2 + CH_3NCH_2$, the barriers are relatively high with 93.5 and 105.8 kcal/mol, respectively. (**e**) Pyrrolidine can also decompose to $CH_3CHCH_2 + CH_2NH$. In this process, the C–C and C–N bond cleavages occur with the concomitant intramolecular hydrogen transfer and a barrier of 95.0 kcal/mol. The CASPT2 results for the pathways (**a**) and (**b**) (including the biradicals) are listed in Table S1. The barriers are higher at the CASPT2 level. However, the pathway (**a**) is still favorable than (**b**), which is consistent with the G3B3 results. Lifshitz et al. have reported that (**a**) and (**e**) are the main pyrrolidine pyrolysis channels [19], consistent with the present calculations and experimental results, where CH_2CH_2, CH_3CHCH_2, and CH_2NH have been identified in the flame.

4.3.2. Reactions of pyrrolidine radicals

The reaction pathways of pyrrolidine radicals (**PC1** and **PC2**) are depicted in Fig. 5 (energies are given relative to **PC1**). **PC1** can isomerize to **PC2** via hydrogen transfer (**TS14**, 40.6 kcal/mol). Further H-elimination of **PC2** can yield **PC7**. Also, **PC2** can open the ring to form **PC13** via α-C–β-C bond cleavage with a transition state of 33.4 kcal/mol above **PC1**. **PC7** can also be produced from **PC1** via N–H bond cleavage. The other dihydro-pyrrole isomer **PC8** results from β-C–H bond cleavage, with dissociation energies of 36.2 kcal/mol. The further α-C–H bond cleavage of **PC1** can form a carbene product **PC10** with dissociation energy of 62.3 kcal/mol. Two ring-opening pathways for **PC1** are proposed. The barrier of the α-C–N bond cleavage reaction is lower with 28.1 kcal/mol (**TS15**). For the β-C–β-C bond cleavage reaction, the barrier is 33.8 kcal/mol, and the ring-opening product **PC12** is less stable.

The unimolecular decomposition and ring-opening pathways from two dihydro-pyrrole isomers (**PC7** and **PC8**) are presented as Supplementary data. Compared to the ring-opening pathway, the dissociation reactions are more energy-costly for the unimolecular reactions of dihydro-pyrrole and the barriers are relatively high. Similar to the decomposition of pyrrolidine, the main possible decomposition pathways are also the radical attack reactions and the subsequent β-scissions.

Based on the experimental measurements and theoretical calculations, Fig. 6 displays the major consumption pathways of pyrrolidine, including the H-loss, ring-opening, H_2-elimination and dissociation channels, and secondary dissociation and ring-opening reactions. Pyrrolidine radical, 3,4-dihydro-pyrrole (**PC7**), 2,3-dihydro-pyrrole (**PC8**) and some small molecules have been

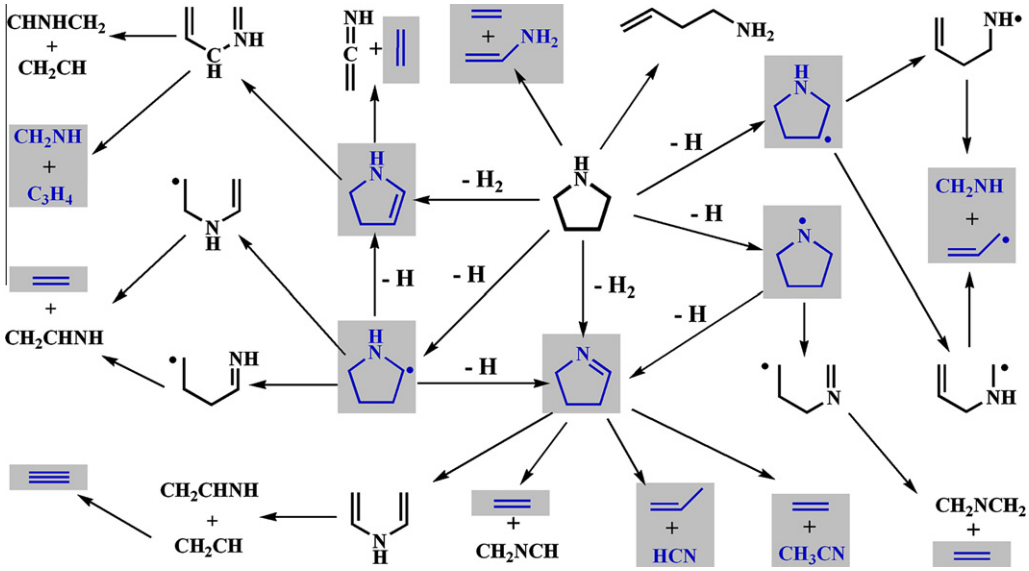

Fig. 6. Main decomposition pathways of pyrrolidine. Detected species are highlighted.

detected in the experiment, which are highlighted in Fig. 6. The detailed theoretical study on the primary and secondary decomposition channels of pyrrolidine can shed light on the major reaction pathways and explain the experimental results. They will be valuable for the development of a detailed kinetic model for pyrrolidine pyrolysis and combustion.

The flame chemistry was compared to the morpholine flame investigated by Lucassen et al. [29], which has similar conditions (C/O = 1:2.44) to that investigated here. In that work, the concentration of C3-species is very low and consequently, aromatics are below the detection limit. Here, various C3–C5 species and aromatics like pyrrole, benzene, pyridine, 2-methylpyrrole, and toluene were detected. This observation could be related to the fuel structure which presents a chain with four C-atoms rather than the C2-elements in morpholine. In the pyrrole flame [9], more high-mass species were identified. However, the pyrrole flame was richer (C:O:N = 1:1.56:0.25) than the present pyrrolidine flame. For a better comparison of the flame chemistry, flames with identical conditions are needed for further study.

5. Conclusions

A fuel-rich laminar premixed low-pressure pyrrolidine flame was investigated with molecular-beam mass spectrometry using tunable vacuum ultraviolet photoionization. A large number of hydrocarbon, oxygenated and nitrogenous combustion intermediates was identified with the help of photoionization efficiency spectra. Pollutant compounds including nitric oxide, ammonia, hydrogen cyanide, acetonitrile, formaldehyde, acetaldehyde, and propanal were detected. Identified intermediates also include some radicals (C_4H_8N, C_4H_7, C_3H_5, C_3H_3, C_2H_5, CHO, and CH_3) and isomers (ethenol and acetaldehyde, propyne and allene, 1,3-butadiene and 1-butyne, and 1-butene and 2-butene). The decomposition and ring-opening pathways of pyrrolidine and the subsequent reactions of the ring decomposition products were analyzed using the G3B3 method, highlighting also H-loss channels. Subsequent reactions of the main H-loss products, pyrrolidine radicals and dihydro-pyrrole, both detected in the experiment, were also proposed. The quantum chemical calculation results can assist in understanding the experimental observations. Combining experimental and theoretical studies, the main primary and secondary decomposition channels of pyrrolidine were discussed. For a deeper understanding of the flame chemistry, comprehensive analysis is required, which includes mole fractions of the obtained species and kinetic modeling.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.034.

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